Aqua-As

Granular iron-hydroxide for effective removal of arsenic from water

General

Due to its chemical activity, iron (III) hydroxide is ideally suitable to bind arsenate ions in aqueous media. However, the technical application of this universal adsorbent failed in the past due to the fact that iron oxide is usually only available as a paste so that the complicated handling procedure prevented its use in filtering columns.

Today, a special process allows the production of iron hydroxide in a granular shape. Moreover, different grain sizes can be produced by grinding and sieving thus leading to many new different fields of application for iron (III) hydroxide.

Due to this important point, various fields of application for the chemical compound **Aqua As** are arising.

Application areas

At present **Aqua As** can be usefully applied for following purposes:

- Separation of arsenate ions in drinking water treatment
- Treatment of contaminated ground water during remediation of past pollution
- Retention of contaminants in “reactive barriers”
- Cleaning industrial waste water polluted with heavy metals

The advantages at a glance

- High cleaning effect at low costs
- High charging performance due to a highly porous surface
- Ecologically favourable due to the use of ferrous side products from waterTreatment when being produced
- Unproblematic and inexpensive waste disposal of the adsorptive agent (the producer guarantees to take back ~~the waste)~~

Effectivity

In the first step, contaminants in the form of arsenate ions dissolved in water are bound by adsorption to the **Aqua As** surface. During the subsequent reaction the trans-formation into stable iron arsenate takes place.

Regarding the ions mechanism involving adsorption with subsequent fixation of the heavy metals in an iron hydroxide “crystal lattice is assumed regarding the binding of heavy metal”.

Furthermore, a quite unspecific binding of dissolved organic water pollutants by way of adsorption is also possible.

The following reaction equation represents the simplified transformation of iron hydroxide with arsenate ions:

$$Fe(OH)_3 + H_3AsO_4 \rightarrow FeAsO_4 + 3\ H_2O$$

Iron arsenate

Technical Data

Materials group	iron hydroxide
Formula	$Fe(OH)_3$
Iron content (Fe^{3+})	min. 40 %
Iron hydroxide content	min. 70 %
Colour	reddish brown
Bulk density	approx. 625 kg/m³ (0,5 – 2,0 mm grain size)
	approx. 600 kg/m³ (2,0 – 4,0 mm grain size)
Density	1,91 g/cm³
Composition	granular
Special surface (BET)	min. 300 m²/g
Total Porosity	min. 70 %
Available grain sizes	0,5 – 2,0 mm
	2,0 – 4,0 mm
Over- und Undersize	each < 5 %
Humidity	< 8 %

Operating parameters

Load capacity	approx. 7 – 9 g As/kg **Aqua As**
Filtration rate	max. 12 – 17 m/h
Residence time	5 to max. 10 bed volumes per hour in case of 2 reactors positioned each after the other, respectively at last in case of serial switching of 2 filters
Filling height intervals	max. 4 – 5 m
Backwash	once / week *

*Note for backwash:

In case of an appropriate filter to remove suspended solids (multi-layer filter) being pre-positioned to the filter to eliminate arsenic, possibly backwashing is not necessary.

Chemical analysis

	Chemical symbol	Content in mg/kg TS
Arsenic	As	< 20
Lead	Pb	< 20
Cadmium	Cd	< 0,5
Chromium	Cr	< 10
Cobalt	Co	< 40
Copper	Cu	< 10
Nickel	Ni	< 35
Mercury	Hg	< 0,05
Zinc	Zn	< 110

All data stated in this specification are average values from several measurements without any legally binding effect.

Note

The information given in the info-brochure is accurately put together, revised and updated if required. However, we cannot be hold responsible for this information as being up-to-date, exact and complete. Apart from that, this information cannot replace a personal consultation in the specific case.